

MATH 3260 - Chapter 8 Worksheets

Complex Vector Spaces

Name: _____

Date: _____

These are student practice worksheets. They are designed to be used with the chapter notebook and notes, but every problem below contains the data needed to work it independently. No solutions are included in this student version.

Section 8.1 - Complex Numbers

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. For $z = 5 - 2i$ and $w = -1 + 4i$, compute $z + w$, $z - w$, $3z - 2w$, and zw .
2. Plot z , w , and $z + w$ in the complex plane and explain the vector-addition interpretation.
3. Solve $x^2 - 8x + 25 = 0$ and verify both roots by substitution.

4. Let

$$A = \begin{bmatrix} 2+i & 1 \\ -i & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2i \\ 2 & 1-i \end{bmatrix}.$$

Compute $A + B$, AB , and $\det(A)$. If $\det(A) \neq 0$, also find A^{-1} .

5. Explain why AB and BA need not be equal even when both products exist.

Section 8.2 - Conjugates and Division

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. For $z = -3 + 4i$, find $\bar{z}$, $z\bar{z}$, and $|z|$.

2. Verify directly that $\overline{z+w} = \bar{z} + \bar{w}$ for $z = 2 - 5i$ and $w = -1 + 3i$.

3. Derive the reciprocal formula

$$\frac{1}{a+bi} = \frac{a-bi}{a^2+b^2}$$

for $a+bi \neq 0$.

4. Simplify

$$\frac{7-3i}{1+2i}$$

by multiplying by the conjugate of the denominator.

5. Prove that $|z/w| = |z|/|w|$ when $w \neq 0$.

6. Find and verify the inverse of

$$A = \begin{bmatrix} 2 & i \\ -i & 1 \end{bmatrix}.$$

Section 8.3 - Polar Form and De Moivre's Theorem

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. Convert $z = -1 - \sqrt{3}i$ to polar form using the principal argument. Then list two other valid arguments of the same complex number.

2. Convert $4 \operatorname{cis}(5\pi/6)$ to rectangular form.

3. Let

$$z_1 = 3 \operatorname{cis}(\pi/3), \quad z_2 = 2 \operatorname{cis}(-\pi/6).$$

Find $z_1 z_2$ and z_1/z_2 in both polar and rectangular forms.

4. Use De Moivre's theorem to compute $(\sqrt{3} + i)^6$.

5. Find all cube roots of -8 and verify one root directly.

6. For each angle below, reduce it to the principal interval $(-\pi, \pi]$:

$$\frac{11\pi}{6}, \quad -\frac{7\pi}{4}, \quad \frac{9\pi}{2}.$$

Explain the difference between an argument and the principal argument.

Section 8.4 - Complex Vector Spaces and Inner Products

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

Convention in this course: $\langle \mathbf{u}, \mathbf{v} \rangle = \sum_j u_j \overline{v_j}$. The inner product is linear in the first argument and conjugate-linear in the second; conjugate symmetry replaces ordinary symmetry.

1. For $\mathbf{u} = (1 + i, 2 - i)$ and $\mathbf{v} = (2, -i)$, compute $\mathbf{u} + \mathbf{v}$ and $(1 - 2i)\mathbf{u}$.
2. Determine whether $\{(1, i), (i, 1)\}$ is a basis for $\mathbb{C}^2$.
3. Compute $\langle \mathbf{u}, \mathbf{v} \rangle$, $\|\mathbf{u}\|$, and $d(\mathbf{u}, \mathbf{v})$ using the vectors in Problem 1.
4. Find the projection of $\mathbf{u} = (1 + i, 2)$ onto $\mathbf{v} = (1, -i)$ and verify that the residual is orthogonal to $\mathbf{v}$.
5. Apply one step of complex Gram-Schmidt to $\mathbf{v}_1 = (1, i)$ and $\mathbf{v}_2 = (1, 1)$.

Section 8.5 - Unitary and Hermitian Matrices

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. Find A^* for

$$A = \begin{bmatrix} 1+i & 2 \\ 3i & -1+i \end{bmatrix}.$$

2. Prove directly that $(cA)^* = \bar{c}A^*$.

3. Determine whether

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ i & -i \end{bmatrix}$$

is unitary. Also verify that its columns form an orthonormal basis of $\mathbb{C}^2$.

4. Verify that

$$H = \begin{bmatrix} 3 & 1+i \\ 1-i & 3 \end{bmatrix}$$

is Hermitian and find its eigenvalues.

5. Unitarily diagonalize

$$H = \begin{bmatrix} 2 & i \\ -i & 2 \end{bmatrix}.$$

6. Using the matrix U from Problem 3 and

$$\mathbf{x} = \begin{bmatrix} 1 \\ i \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 2-i \\ 1 \end{bmatrix},$$

verify numerically that

$$\langle U\mathbf{x}, U\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle.$$